RANKUNO AI AUTOMATION INFRASTRUCTURE

0 → 1 Master Architectural Blueprint & Executive Proposal

A comprehensive engineering plan for transitioning Rankuno's manual SEO, PPC, and Market Research operations into an autonomous AI-driven platform.

Executive Document Summary

This document presents the complete architectural specification, SDLC governance framework, and detailed Phase 1 execution plan for Rankuno's AI Automation Infrastructure. Designed to establish enterprise standards from day zero, this proposal details how Rankuno will automate website classification, technical SEO audits, keyword clustering, and multi-channel intelligence.

Prepared By: Founding AI Automation Lead

Target Domain: SEO Operations Engine (Phase 1)

Architecture: Google Antigravity Agentic Stack

Status: Pending Leadership Review & Sign-off

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1. Executive Vision & Strategic Objectives

Currently, digital marketing operations across **SEO Analysis**, **Paid Search (PPC)**, and **Market Research** are executed manually. While effective, manual workflows do not scale linearly and introduce human error, inconsistent reporting, and operational bottlenecks.

The **Rankuno AI Automation Platform** is designed to build a 0-to-1 enterprise-grade automation infrastructure. The core strategic goals are:

- **10x Scale without Overhead**: Automate repetitive data collection, website structure auditing, and content classification.
- **Zero-Legacy Standards**: Build modular, strongly-typed Python micro-modules with zero technical debt from Day 1.
- **Human-in-the-Loop (HITL) Safety**: Enforce strict human approval controls for high-risk write operations (publishing changes, modifying ad budgets, executing database writes).

2. Google Antigravity Agentic Stack Architecture

The platform is engineered around the **Google Antigravity Agentic Stack**, organizing system components into 7 clean operational layers:

Layer Name	System Component	Technical Implementation	Operational Function in Rankuno
Reasoning Model	LLM Brain	Gemini 3.6 Flash / Pro	Processes natural language instructions, analyzes edge cases, and synthesizes reports.
Agent Loop	Decision Orchestrator	Antigravity Core Loop	Runs autonomous `Plan → Act → Observe → Reason` execution loops.
Subagents	Specialized Workers	`invoke_subagent` API	Spawns isolated background agents for research, refactoring, and test suite verification.
Skills	Procedural Knowledge	`skills/<name>/SKILL.md`	Houses step-by-step operational standard workflows (e.g. `antigravity-sdlc-flow`).
Tools	Native Capabilities	System Tool Interfaces	Atomic execution hands: `read_file`, `write_file`, `run_command`, `grep_search`.
MCP Connectors	External APIs	Model Context Protocol	Connects platform to Google Search Console API, Google Ads API, Ahrefs, and SERP APIs.
Guardrails	HITL & Permissions	Permission Controls	Prevents unauthorized live site modifications or unapproved API financial spend.

3. Mandatory 8-Step SDLC Execution Protocol

Every feature, module, or update developed for Rankuno MUST adhere to this strict 8-step quality loop:

1. **Step 1: Investigation & Requirement Discovery** — Audit domain requirements, existing dependencies, and missing inputs.
2. **Step 2: System Architecture & Data Schema Blueprint** — Define Pydantic models (`BaseModel`) and component boundaries.
3. **Step 3: Human-in-the-Loop (HITL) Review Checkpoint** — Present architecture to operators/leads for explicit sign-off.
4. **Step 4: Reasoning-Driven Implementation Plan** — Detail line-by-line file changes and component logic.
5. **Step 5: Security, Rate-Limit & Edge-Case Audit** — Audit for anti-bot blocks, rate limits, API spend, and breaking changes.
6. **Step 6: Step-by-Step Modular Implementation** — Write clean, typed, modular code adhering to engineering standards.
7. **Step 7: Automated Verification & Testing** — Execute automated unit test suites (pytest); never declare success without passing tests.
8. **Step 8: README & Architecture Drift Audit** — Immediately update `README.md` and `docs/ARCHITECTURE.md` to stay in sync.

4. Phase 1 Master Plan: Multi-Dimensional Page & URL Classification Engine

The first production module to be constructed is the **Page & URL Classification Engine**. It classifies any website's URLs across **3 complementary interfaces** to construct a complete, structured content graph.

```
                      Target Website URL / Crawled DOM
                                  |
            +---------------------+---------------------+
            v                     v                     v
+-------------------------+-------------------------+-------------------------+
|      INTERFACE 1:       |      INTERFACE 2:       |      INTERFACE 3:       |
| Page Type & Hierarchy   | Theme & Topical Cluster | Semantic Intent &       |
| Classification          | Classification          | Conversion Role         |
+-------------------------+-------------------------+-------------------------+
| - L0 Homepage           | - Business Niche        | - Informational         |
| - L1 Primary Nav Hub    | - Topical Silo          | - Commercial Search     |
| - L2 Sub-Nav Hub        | - Sub-Niche Product     | - Transactional         |
| - L3 Leaf Money Page    | - Entity Mapping        | - Navigational          |
| - Utility Pages         |                         | - Lead Gen vs Sale      |
+-------------------------+-------------------------+-------------------------+
```

4.1 Interface 1: Hierarchy Taxonomy & Functional Page Types

Hierarchy Level	Level Name	Primary Page Types Included	Example URL Pattern
`L0_HOMEPAGE`	Root Entry	Homepage	`https://vitaquest.com/`
`L1_PRIMARY_NAV_HUB`	Global Parent Hub	Service Category Hub, Blog Hub, Company Hub, Lead Gen Hub	`/manufacturing-services/` , `/capabilities/`
`L2_SUB_NAV_HUB`	Sub-Category Hub	Sub-Service Category, Product Category Collection, Tag Archive	`/manufacturing-services/capsule-supplements/`
`L3_LEAF_PAGE`	Money / Execution Page	Service Detail Page (SDP), Product SKU (PDP), Blog Article	`/blog/supplement-manufacturing-trends`
`UTILITY_PAGE`	Infrastructure	Privacy Policy, Terms, Search Results, Faceted Filters, 404s	`/privacy-policy` , `/shop?color=red`

4.2 The 6-Signal Consensus Architecture (Zero-Failure Engine)

Traditional SEO scrapers fail when websites use hidden mobile menus, flat URL structures (e.g. `site.com/capsules` off root), or dynamic JavaScript. Rankuno solves this by combining **6 independent signals** into a weighted consensus engine:

1. **Signal 1: ARIA Navigation Tree Analysis** — Inspects DOM accessibility attributes (`role="navigation"` , `aria-label="Main menu"`) and nested `<ul>/<li>` HTML parent-child structures. This accurately classifies Level 1 vs. Level 2 navigation items even if CSS hides them behind a **Hamburger toggle** (`display: none`).
2. **Signal 2: CMS API Endpoint Footprints** — Queries public CMS data endpoints (WordPress `/wp-json/wp/v2/pages` , Shopify `/collections.json` , `/products.json`). Instantly provides 100% accurate classification for **Flat URLs**.

3. **Signal 3: Grouped XML Sitemap Index Parsing** — Parses `sitemap_index.xml` to inspect auto-grouped files (e.g. `product-sitemap.xml`, `category-sitemap.xml`, `post-sitemap.xml`).
4. **Signal 4: Schema.org `@graph` JSON-LD Parsing** — Extracts structured schema types embedded in HTML: `CollectionPage`, `ItemPage`, `Service`, `Product`, `Article`, `AboutPage`.
5. **Signal 5: Internal Link In-Degree Centrality** — Evaluates internal link distribution. Pages linked in site-wide headers/footers (≥ 1,000 links) are classified as Level 1 Hubs.
6. **Signal 6: LLM Zero-Shot Backup Classifier** — Passes URL, Title, H1, Breadcrumbs, and text snippets to Gemini 3.6 Flash for intelligent fallback classification when zero structural signals exist.

5. Pydantic Data Contracts & Schemas

All data flowing through the classification engine is strictly typed using Pydantic v2 data models:

```python
from enum import Enum
from typing import List, Optional
from pydantic import BaseModel, Field

class HierarchyLevel(str, Enum):
    L0_HOMEPAGE = "L0_HOMEPAGE"
    L1_PRIMARY_NAV_HUB = "L1_PRIMARY_NAV_HUB"
    L2_SUB_NAV_HUB = "L2_SUB_NAV_HUB"
    L3_LEAF_PAGE = "L3_LEAF_PAGE"
    UTILITY_PAGE = "UTILITY_PAGE"

class PrimaryPageType(str, Enum):
    HOMEPAGE = "HOMEPAGE"
    SERVICE_CATEGORY_HUB = "SERVICE_CATEGORY_HUB"
    SERVICE_DETAIL_PAGE = "SERVICE_DETAIL_PAGE"
    PRODUCT_CATEGORY_HUB = "PRODUCT_CATEGORY_HUB"
    PRODUCT_DETAIL_PAGE = "PRODUCT_DETAIL_PAGE"
    BLOG_HUB = "BLOG_HUB"
    BLOG_ARTICLE = "BLOG_ARTICLE"
    COMPANY_ABOUT = "COMPANY_ABOUT"
    COMMERCIAL_LEAD_GEN = "COMMERCIAL_LEAD_GEN"
    FACETED_FILTER = "FACETED_FILTER"
    UTILITY_LEGAL = "UTILITY_LEGAL"
    UNKNOWN = "UNKNOWN"

class SearchIntent(str, Enum):
    INFORMATIONAL = "INFORMATIONAL"
    COMMERCIAL_INVESTIGATION = "COMMERCIAL_INVESTIGATION"
    TRANSACTIONAL = "TRANSACTIONAL"
    NAVIGATIONAL = "NAVIGATIONAL"

class SignalSource(str, Enum):
    SITEMAP_INDEX = "SITEMAP_INDEX"
    ARIA_NAV_TREE = "ARIA_NAV_TREE"
    SCHEMA_JSONLD = "SCHEMA_JSONLD"
    CMS_API_ENDPOINT = "CMS_API_ENDPOINT"
    LINK_IN_DEGREE = "LINK_IN_DEGREE"
    LLM_ZERO_SHOT = "LLM_ZERO_SHOT"

class SignalScore(BaseModel):
    source: SignalSource
    suggested_level: HierarchyLevel
    suggested_page_type: PrimaryPageType
    confidence: float = Field(ge=0.0, le=1.0)
    notes: str

class FullPageIntelligenceProfile(BaseModel):
    url: str
    normalized_path: str
    hierarchy_level: HierarchyLevel
    primary_page_type: PrimaryPageType
    nav_parent_url: Optional[str]
    breadcrumb_path: List[str]
    topical_category: str
    sub_topic: Optional[str]
    search_intent: SearchIntent
    conversion_role: str
    signals_evaluated: List[SignalScore]
    final_confidence_score: float = Field(ge=0.0, le=1.0)
    consensus_method: str
```

6. Edge-Case, Security & Risk Mitigation Audit

Edge Case / Technical Vulnerability	Root Cause / Impact	Rankuno Architectural Guardrail
Hamburger / Hidden Mobile Navs	CSS hides navigation (`display: none`).	Parse DOM ARIA tree (`role="navigation"`) ignoring visual CSS display.
Flat URLs (`site.com/capsules`)	Directory path depth regex fails.	Query CMS Endpoints (`/collections.json` , `/wp-json/`) & Sitemap index files.
JavaScript SPAs (React/Next.js)	Empty `<div id="root">` returned.	Playwright headless browser rendering fallback to hydrate dynamic DOM.
Multi-Language Routing (`/en/` , `/es/`)	Locale prefixes skew category matching.	Path Normalizer: Strip locale prefixes (`/en/services` → `/services`) prior to matching.
Faceted Filter Query Params	Filter parameters (`?color=red`) create infinite URLs.	Parameter Normalizer: Group query parameter URLs into `FACETED_FILTER` utility type.
Anti-Bot & Cloudflare Rate Limits	IP blocks during bulk scraping.	Rotate User-Agent headers, enforce backoff retries, and prefer CMS API endpoints.
API Cost Overflow	Unbounded LLM fallback API calls.	Rate-limiter & caching layer: LLM invoked only when 5 structural signals are ambiguous.

7. Multi-Phase Strategic Implementation Roadmap

``` Phase 1: Multi-Dimensional Page Classification Engine [CURRENT PHASE] ├── Phase 1A: Blueprint, Schemas & Edge-Case Audit (Sign-off) ├── Phase 1B: Core Signal Parsers (ARIA, CMS Client, Sitemap, Schema) ├── Phase 1C: Consensus Engine & LLM Zero-Shot Backup └── Phase 1D: Automated Verification & Unit Test Suite Phase 2: Technical SEO Crawling & Audit Engine ├── Dynamic JS Crawler & DOM Parser (Playwright) ├── Core Web Vitals (PageSpeed Insights API Integration) └── Automated Technical Audit Reporter Phase 3: Keyword Intelligence & Vector Clustering Engine ├── Semantic Embeddings & Intent Classifier └── Content Gap & Cannibalization Analyzer Phase 4: Paid Search (PPC) Campaign & Ad Engine ├── Automated RSA Copy Generator & Negative Keyword Finder └── PPC Performance Anomaly Sentinel Phase 5: Autonomous Multi-Agent Orchestration & Command Dashboard ├── Multi-Agent Execution (SEO Agent + PPC Agent + Research Agent) └── Executive Analytics & Web Command UI ```

8. Leadership Approval & Sign-Off Block

> **Approval Request:** By signing below, Rankuno leadership approves the Architectural Blueprint, SDLC Governance Protocol, and Phase 1 Implementation Plan for the Rankuno AI Automation Infrastructure.

Executive Sign-Off Form

Submitted By:	**Approved By (Leadership / Executive):**
______________________________	______________________________
Founding AI Automation Lead	Executive Sponsor / CTO
Date: July 29, 2026	Date: ______________
```